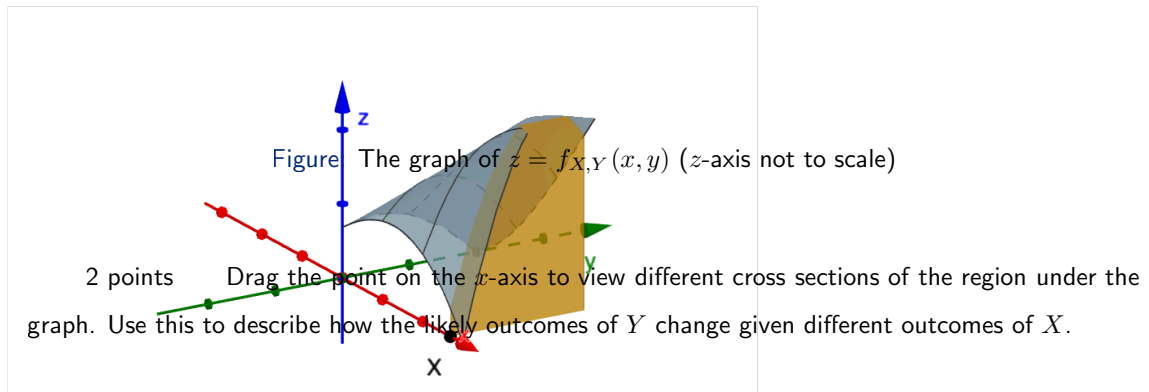


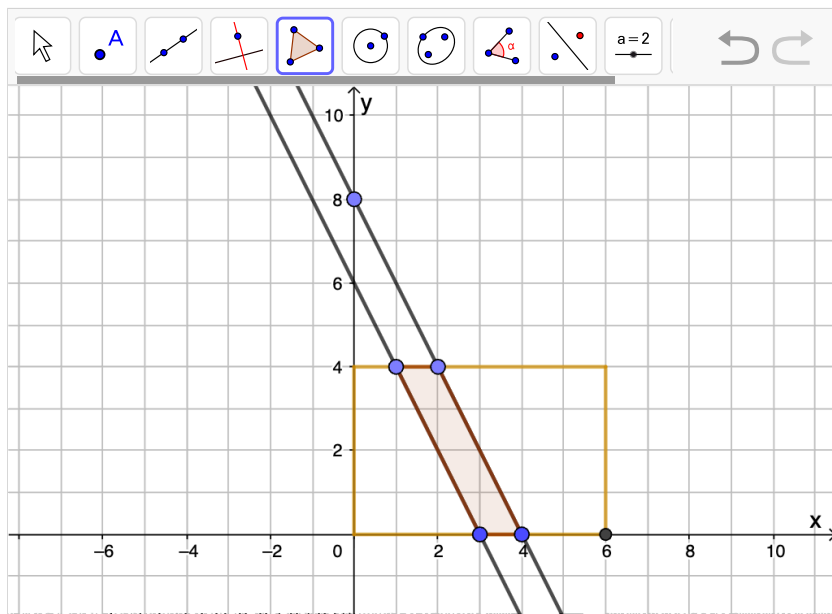


The area of the cross-section of the graph increases as X , which means the likely outcomes of Y is associated with higher values of X .

A1

Functions of Joint Random Variables

2 points Consider the function $2X + Y$. Suppose we wanted to compute the probability that $2X + Y$ is between 6 and 8. Draw the domain that we would integrate $f_{X,Y}(x, y)$ over. Then set up the integral.



$$6 \leq 2X + Y \leq 8$$

$$y = -2x + 6, y = -2x + 8$$

$0 < y < 4$ 2 points Write an integral that would compute $E[2X + Y]$. Do not evaluate it.

$$P(6 \leq 2X + Y \leq 8) = \int_{y=0}^{y=4} \int_{x=\frac{(6-y)}{2}}^{x=\frac{(8-y)}{2}} f(x,y) dx dy$$



$$E[2X + Y] = \int_{y=0}^{y=4} \int_{x=\frac{(6-y)}{2}}^{x=\frac{(8-y)}{2}} (2x + y) \left(\frac{(x+2)(81 - (3y-3-x)^2)}{2640} \right) dx dy$$

A2

Functions of Joint Random Variables

2 points Based on your answer to , how is $E[2X + Y]$ related to $E[X] = 3.58$ and $E[Y] = 2.07$? What rules are you using to come to that conclusion?



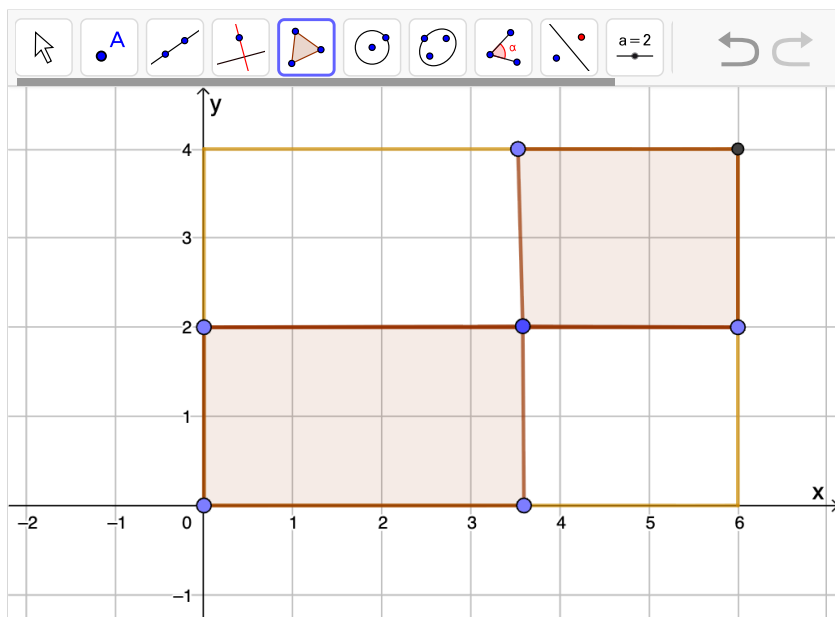
If $E[X] = 3.58$ and $E[Y] = 2.07$, we can assume that $E[2X+Y] = 2E[X] + E[Y] = 2(3.58) + 2.07 = 9.23$.



A3

Functions of Joint Random Variables

2 points Consider the function $(X - 3.58)(Y - 2.07)$. Draw one or more shapes on the domain below to indicate the region where this function is positive. If most of the outcomes are in the region you indicated, how would you describe the relationship between X and Y ?



For the function to be positive, we need both X and Y to have the same sign, either positive positive, or negative negative. Therefore, the regions cover areas where $X < 3.58$ and $Y < 2.07$ and where $X > 3.58$ and $Y > 2.07$.

From this, we can assume that if X has an expected value that is smaller than 3.58, we can assume that the expected value of Y is smaller than 2.07. Similarly, if X has an expected value bigger than 3.58, we can assume that the expected value of Y is bigger than 2.07

When you are done, make sure your graphs show the information you want to submit, then

[click to make a pdf](#)

If that doesn't work, use your browser's print function and save as a pdf

A4